

Lab Manual
Course Code: ETE 3212
Course Title: Sessional Based on ETE 3211

No.	Name of the experiments
1.	Computation of Birefringence of elliptical shaped core optical fiber using COMSOL Multiphysics Software.
2.	Computation of Confinement Loss for Circular core optical fiber using COMSOL Multiphysics Software.
3.	Experimental analysis of analog/digital/voice communication using fiber optics cable.
4.	Experimental analysis of measuring the numerical aperture (NA) of optical fiber.
5.	Experimental analysis of measuring the propagation loss or attenuation loss in optical fiber.
6.	Experimental analysis of measuring the bending loss in Optical fiber.

Experiment 1: Computation of Birefringence of Elliptical-Shaped Core Optical Fiber using COMSOL Multiphysics Software

Objective: To model an elliptical-core optical fiber using COMSOL Multiphysics, compute the effective refractive indices of orthogonal polarization modes, and evaluate the birefringence of the fiber.

Theory:

Optical Fiber Birefringence:

In a perfectly circular-core fiber, two orthogonal polarizations (E_x and E_y) have the same propagation constant. However, in an elliptical-core fiber, the asymmetry leads to different propagation constants, resulting in birefringence.

Birefringence (B):

$$B = |n_x - n_y|$$

where n_x and n_y are the effective refractive indices of orthogonal modes.

Applications: Polarization-maintaining fibers, sensors, modulators, and devices requiring stable polarization.

Software & Tools Required:

- COMSOL Multiphysics (Wave Optics Module)
- Computer with sufficient RAM (≥ 8 GB recommended)
- Fiber design parameters

Design Parameters:

Typical parameters for simulation (values can be changed as per requirement):

- Core shape: Elliptical (major axis a , minor axis b)
- Core refractive index (n_{core}) = 1.450
- Cladding refractive index (n_{clad}) = 1.444
- Wavelength (λ) = 1.55 μm
- Core size: $a = 4 \mu\text{m}$, $b = 3 \mu\text{m}$
- Cladding radius: 25 μm

Experimental Procedure:

Open COMSOL: Launch COMSOL Multiphysics and select Wave Optics Module $\rightarrow$ Mode Analysis (2D).

Define Geometry: Draw two concentric domains – ellipse for the core (a , b) and circle for cladding (radius R).

Assign Materials: Core with refractive index n_{core} , cladding with n_{clad} . Set boundary as PML or scattering.

Physics Setup: Choose Electromagnetic Waves, Frequency Domain (ewfd). Define $\lambda = 1.55 \mu\text{m}$. Apply scattering boundary condition.

Mesh Generation: Use fine mesh in core region and coarser mesh outside cladding.

Mode Analysis: Add Mode Analysis study, compute eigenmodes, extract effective indices (n_{eff}).
Calculation of Birefringence: Record n_x and n_y , compute $B = |n_x - n_y|$.
Visualization: Plot mode field distributions (E_x and E_y).

Observations: Effective mode index of X- and Y-polarization.

Result:

The birefringence of the elliptical-core fiber is successfully computed using COMSOL Multiphysics. The difference in effective indices of E_x and E_y polarizations demonstrates polarization-dependent propagation.

Questions

What is birefringence in optical fibers?
Why does an elliptical core introduce birefringence?
How do you differentiate between form birefringence and stress birefringence?
What is the significance of polarization-maintaining fibers?
Which COMSOL module is best suited for fiber birefringence analysis?

Experiment 2: Computation of Confinement Loss for Circular Core Optical Fiber using COMSOL Multiphysics Software.

Objective: To model a circular-core optical fiber using COMSOL Multiphysics, compute the effective refractive index of the guided mode, and evaluate the confinement loss of the fiber.

Theory:

Confinement Loss:

In optical fibers, confinement loss arises when part of the guided mode leaks into the cladding or surrounding medium. For circular-core fibers, this loss is typically very small but can become significant in microstructured or leaky fibers.

Confinement Loss (L^c) is expressed as:

$$L^c \text{ (dB/m)} = (40\pi/\ln(10)) \times (\text{Im}(n_{\text{eff}})/\lambda)$$

where $\text{Im}(n_{\text{eff}})$ is the imaginary part of the effective refractive index and λ is the wavelength.

Applications: Fiber design optimization, photonic crystal fibers, and specialty fibers.

Software & Tools:

- COMSOL Multiphysics (Wave Optics Module)
- Computer with sufficient RAM (≥ 8 GB recommended)
- Fiber design parameters

Design Parameters:

Typical parameters for simulation (values can be changed as per requirement):

- Core shape: Circular
- Core refractive index (n_{core}) = 1.450
- Cladding refractive index (n_{clad}) = 1.444
- Wavelength (λ) = 1.55 μm
- Core radius: 4 μm
- Cladding radius: 25 μm

Experimental Procedure:

Open COMSOL: Launch COMSOL Multiphysics and select Wave Optics Module → Mode Analysis (2D).

Define Geometry: Draw two concentric circles – core (radius r_{core}) and cladding (radius r_{clad}).

Assign Materials: Core with refractive index n_{core} , cladding with n_{clad} . Set outer boundary as PML (Perfectly Matched Layer).

Physics Setup: Choose Electromagnetic Waves, Frequency Domain (ewfd). Define $\lambda = 1.55 \mu\text{m}$. Apply scattering boundary condition.

Mesh Generation: Use fine mesh in core region and coarser mesh outside cladding.

Mode Analysis: Add Mode Analysis study, compute eigenmodes, extract effective index (n_{eff}).
Calculation of Confinement Loss: From the complex effective index ($n_{\text{eff}} = n_r + jn_i$), compute confinement loss using:

$$L^c \text{ (dB/m)} = (40\pi/\ln(10)) \times (\text{Im}(n_{\text{eff}})/\lambda).$$

Visualization: Plot mode field distribution to observe confinement in the core.

Observations: Observe n_{eff} .

Result:

The confinement loss of the circular-core fiber is successfully computed using COMSOL Multiphysics. The imaginary part of the effective refractive index provides the leakage loss, confirming light confinement properties.

Questions

What is confinement loss in optical fibers?

Why is confinement loss negligible in standard fibers but significant in microstructured fibers?

How is confinement loss related to the imaginary part of the effective index?

What is the role of PML in confinement loss computation?

Which COMSOL module is used for this experiment?

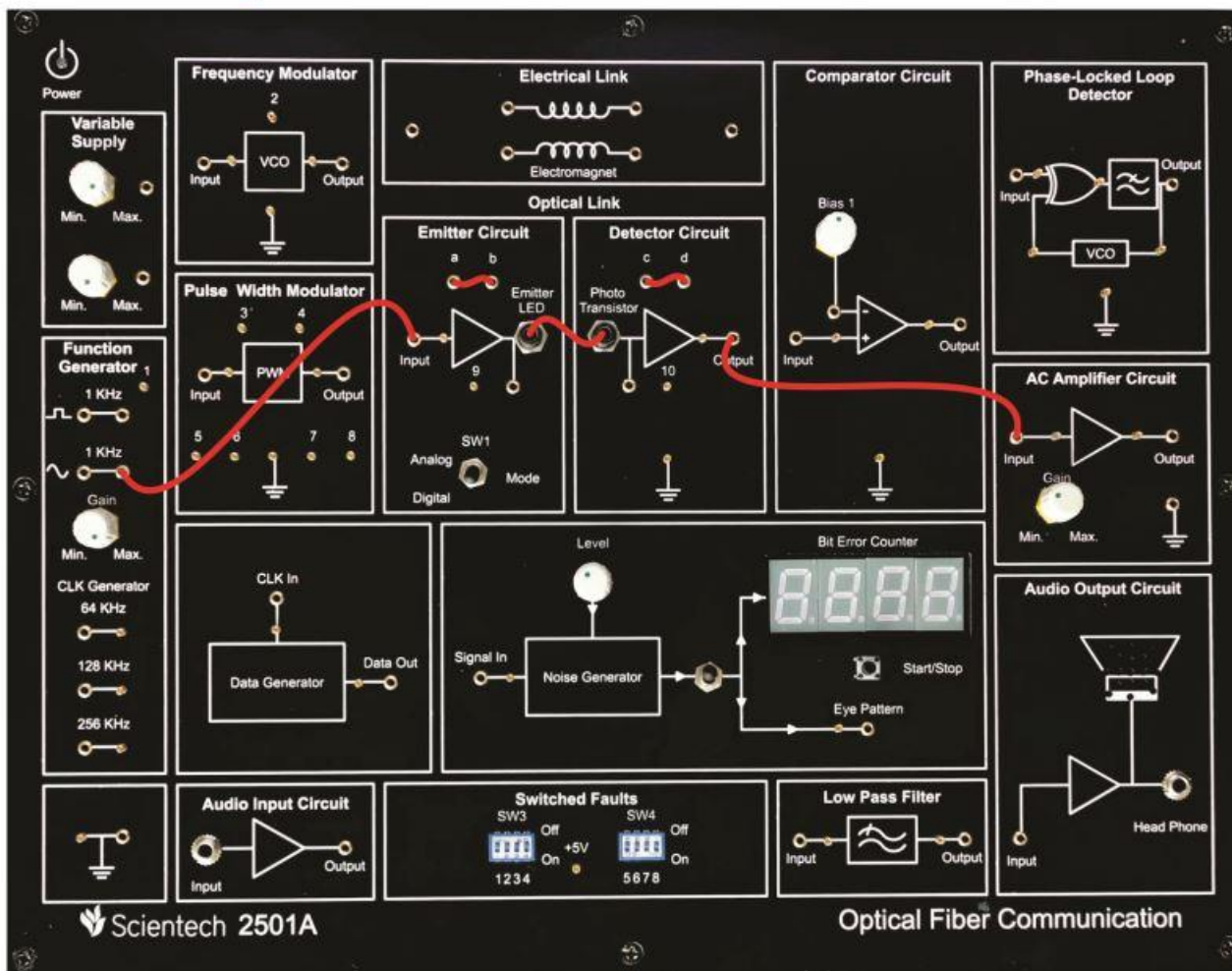
Experiment 3: Experimental analysis of analog/digital/voice communication using fiber optics cable.

Objective: Study of 650 nm Fiber Optic Analog link. In this experiment you will study the relationship between the input signal and received signal.

Equipments Required:

- Sciencetech 2501A TechBook with Power Supply cord
- Optical Fiber cable
- Sciencetech Oscilloscope with necessary connecting probe

Connection Diagram:

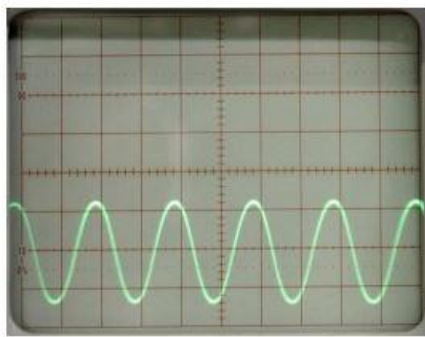


Procedure:

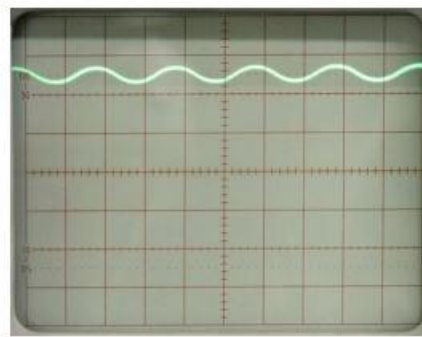
1. Connect the TechBook Power Supply with mains cord to TechBook Sciencetech 2501A.

2. Ensure that all switched faults are OFF. Make the connections as shown in next figure. Connect the Function Generator 1 KHz sine wave output to emitter input.
3. Connect the fiber optic cable between emitter output and detector input.
4. Connect the detector output to AC amplifier input.
5. Put the mode switch SW1 to Analog to drive the emitter in analog mode.
6. Switch 'On' the Power Supply of the TechBook and Oscilloscope.
7. Observe the input to emitter with the output from AC amplifier on Oscilloscope.

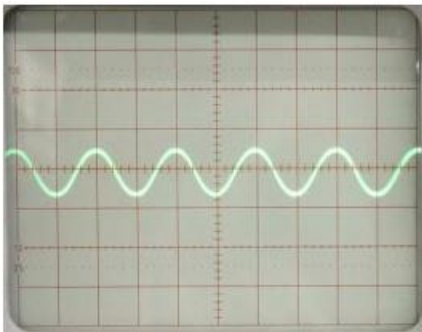
Observation: Observe the waveforms as shown below. Both the input and output waveforms are same.



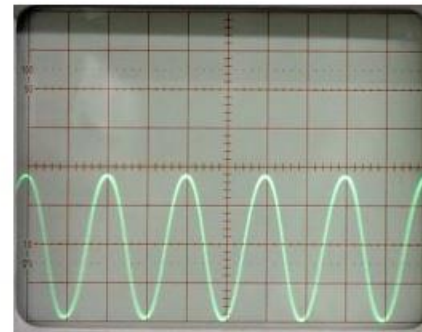
Input to Emitter circuit
(CH1-2V/div; TB- 0.5mS)



Input to Emitter LED
(CH1-2V/div; TB- 0.5mS)



Output of Detector circuit
(CH1-0.2V/div; TB- 0.5mS)



Output of AC Amplifier circuit
(CH1-2V/div; TB- 0.5mS)

Questions:

- What is meant by index profile?
- What is the drawback of multimode Fibers?
- What is Fiber optics?

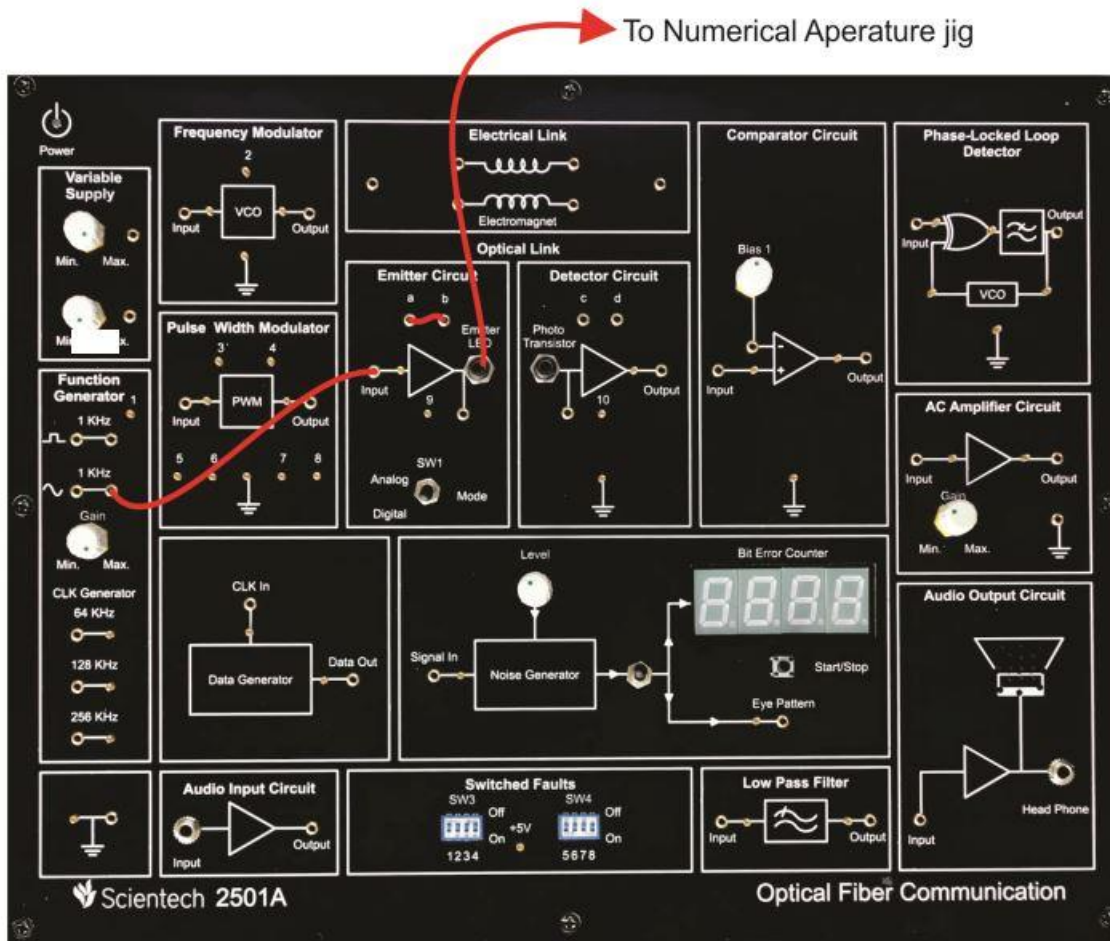
Experiment 4: Experimental analysis of measuring the numerical aperture (NA) of optical fiber.

Objective: Measurement of Numerical Aperture (NA) of optical fiber

Equipments Required:

- Sciencetech 2501A TechBook with Power Supply and mains cord
- Optical Fibre cable
- Numerical Aperture measurement Jig

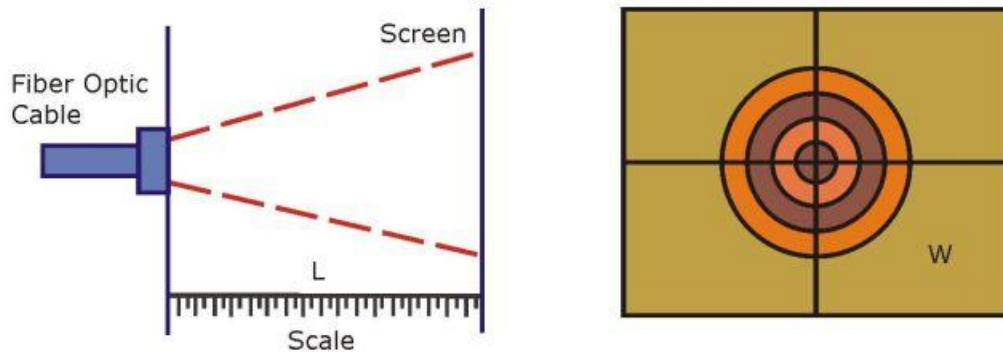
Connection Diagram:



Procedure:

1. Connect the TechBook Power Supply with mains cord to TechBook Sciencetech 2501A.
2. Put the mode switch SW1 to Digital to drive the emitter in digital mode.
3. Switch 'On' the Power Supply of TechBook and Oscilloscope.
4. Connect the Frequency Generator 1 KHz sine wave output to input of emitter circuit. Adjust its amplitude at 5Vp-p.
5. Connect one end of fiber cable to the output socket of emitter circuit and the other end to the numerical aperture measurement jig. Hold the white screen facing the fiber such that its cut face is perpendicular to the axis of the fiber.

6. Hold the white screen with 4 concentric circles (10, 15, 20 & 25 mm diameter) vertically at a suitable distance to make the red spot from the fiber coincide with 10 mm circle



7. Record the distances of screen from the fiber end L and note the diameter W of the spot.
8. Compute the numerical aperture.
9. Vary the distance between in screen and fiber optic cable and make it coincide with one of the concentric circles. Note its distance.
10. Tabulate the various distances and diameter of the circles made on the white screen and computer the numerical aperture from the formula given above.

Inferences: The numerical aperture as recorded in the manufacturer's data sheet is 0.5 typically. The value measured here is 0.35. The lower reading recorded is mainly due to the fiber being under filled.

Questions:

- What is numerical aperture?
- Write the formula for numerical aperture?
- What is the significance of numerical aperture?

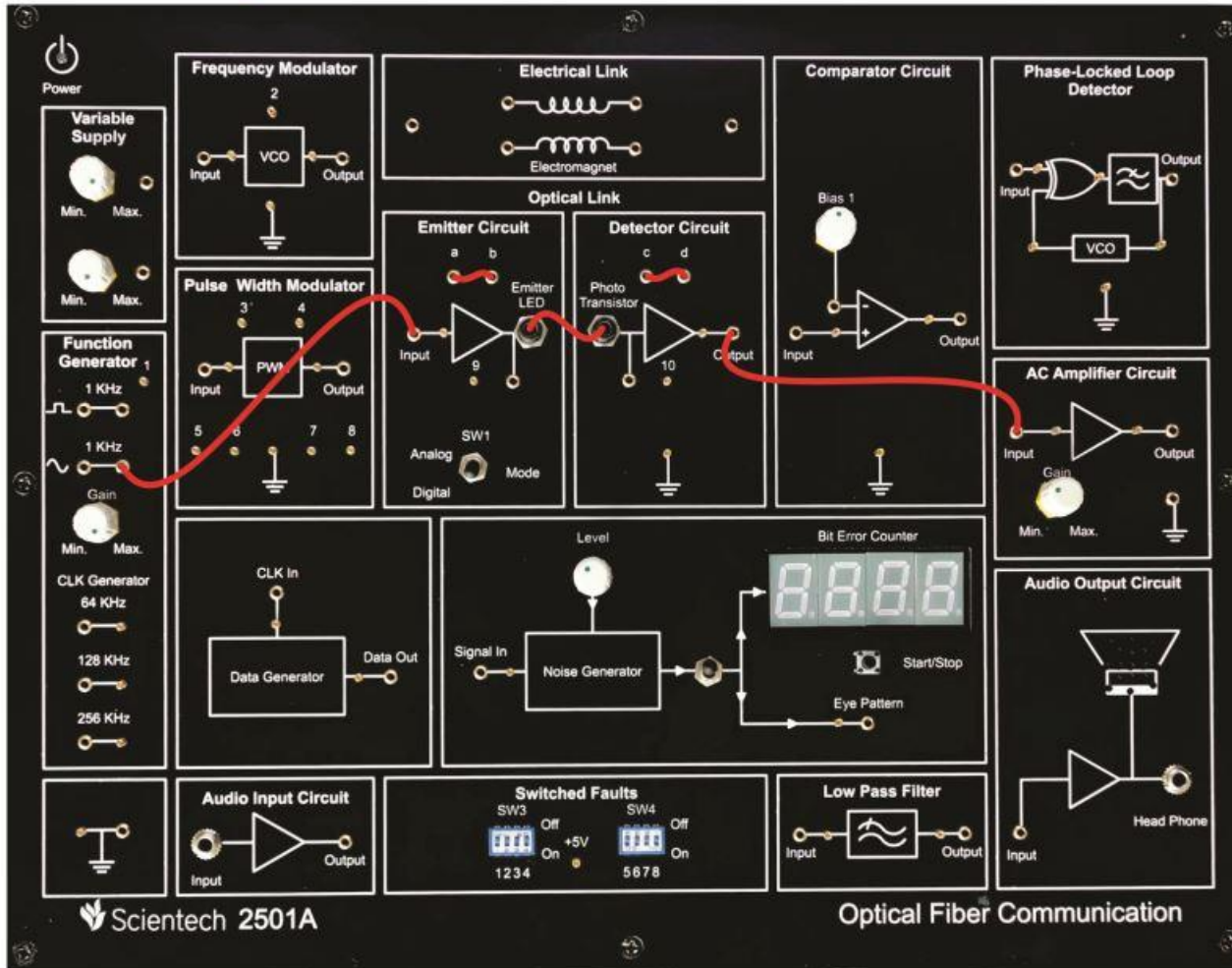
Experiment 5: Experimental analysis of measuring the propagation loss or attenuation loss in optical fiber.

Objective: Measurement of Propagation or Attenuation Loss in the optical fiber

Equipments Required:

- Sciencetech 2501A TechBook with Power Supply Cord
- Optical Fiber Cable
- Sciencetech Oscilloscope with necessary connecting probe

Connection Diagram:



Procedure:

1. Connect the TechBook Power Supply with mains cord to TechBook Sciencetech 2501A.
2. Make the connections as shown in next figure
3. Connect the Function Generator 1 KHz sine wave output to emitter input.
4. Connect 0.5 m optic fiber between emitter output and detector input.
5. Connect Detector output to amplifier input.
6. Put the mode switch SW1 to Analog to drive the emitter in analog mode. Switch 'On' the Power Supply of TechBook and Oscilloscope.

7. Set the Oscilloscope channel 1 to 0.5 V/ Div and adjust 4-6 div amplitude by using X 1 probe with the help of variable potentiometer in Function Generator block at input of emitter.
8. Observe the output signal from detector on Oscilloscope.
9. Adjust the amplitude of the received signal as that of transmitted one with the help of gain adjusts pot in AC amplifier block. Note this amplitude and name it V_1 .
10. Now replace the previous fiber optic cable with 1 m cable without disturbing any previous setting.
11. Measure the amplitude at the receiver side again at output of amplifier. Note this value and name it V_2 . Calculate the propagation (attenuation) loss with the help of following formula.

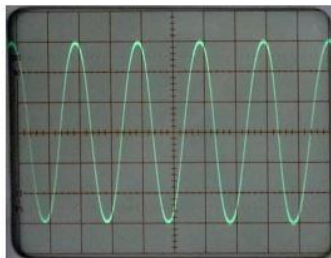
$$\frac{V_1}{V_2} e^{-\alpha(L_1+L_2)}$$

Where α = loss in nepers / meter 1 nepers = 8.686 dB

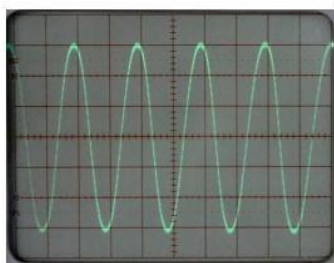
L_1 = length of shorter cable (0.5 m)

L_2 = Length of longer cable (1 m)

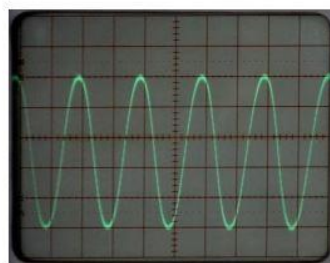
Observations: Observe the waveforms as shown below.



Input to Emitter Circuit (CH1-0.2V/div; TB- 0.5mS)



Detector output (for 0.5m Cable)
(CH1-0.1V/div; TB- 0.5mS)



Detector output (for 1m Cable)
(CH1-0.1V/div; TB- 0.5mS)

Calculation: Recorded values are as follows.

$$\begin{aligned} V_1 &= 0.6V; & V_2 &= 0.54V; \\ \Rightarrow & & \alpha &= - [\log (V_1 / V_2)] / (L_1 + L_2) \\ &= - (0.18/1.5) & &= (- 0.12) \text{ nepers / meter} \\ &= - (0.12) * 8.686 \text{ dB} & &= -1.0 \text{ dB approx.} \end{aligned}$$

$$L_1 = 0.5\text{m}; \quad L_2 = 1.0\text{m}$$

Questions:

- How to measure propagation losses?
- By what optical cable is made up of?
- What is step index Fiber?

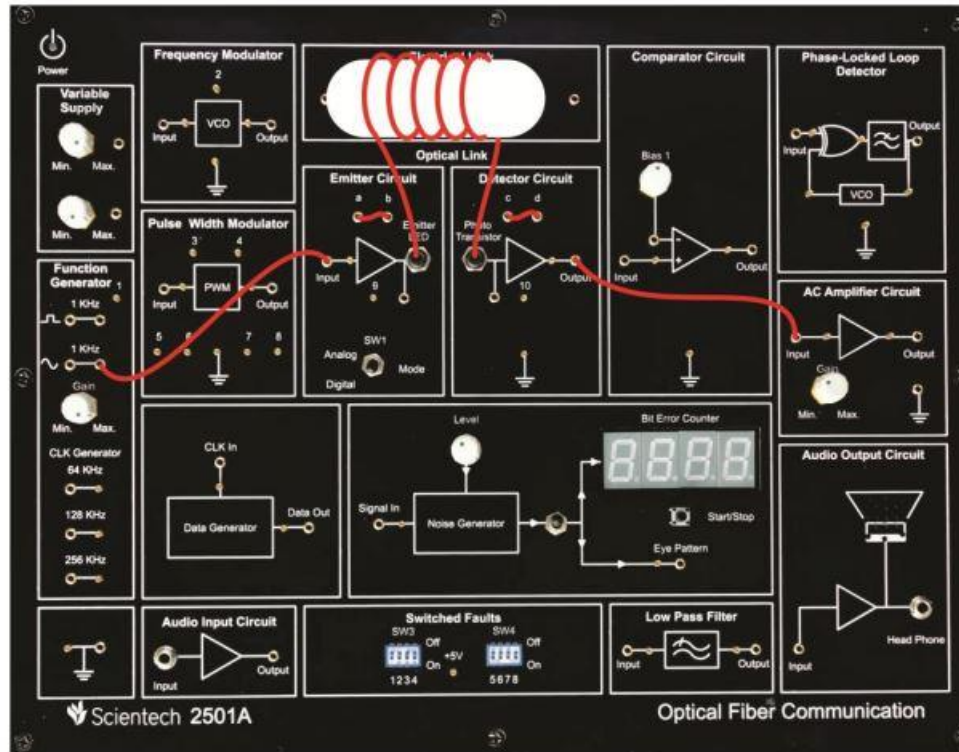
Experiment 6: Experimental analysis of measuring the bending loss in Optical fiber.

Objective: Study of Bending Loss

Equipments Required:

- Sciencetech 2501A TechBook with Power Supply cord
- Optical Fiber cable
- Sciencetech Oscilloscope with necessary connecting probe
- Mandrel

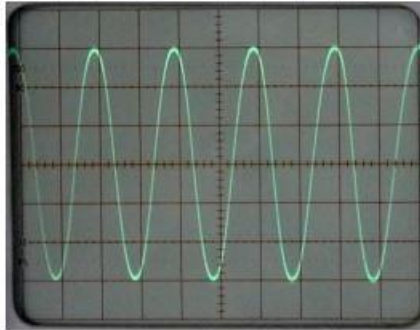
Connection Diagram:



Procedure:

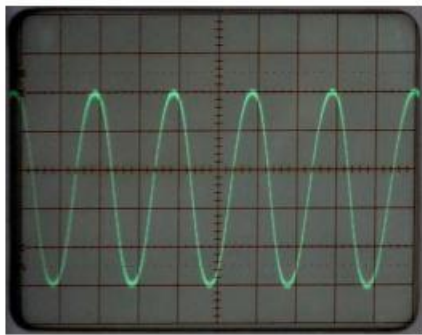
1. Connect the TechBook Power Supply with mains cord to TechBook Sciencetech 2501A.
2. Make the connections as shown in next figure
3. Connect Function Generator 1 KHz sine wave output to emitter input.
4. Connect 0.5 m optic fiber between emitter output and detectors input.
5. Connect Detector output to amplifier input.
6. Put the mode switch SW1 to Analog to drive the emitter in analog mode. Switch 'On' the Power Supply of the TechBook and Oscilloscope.
7. Set the Oscilloscope channel 1 to 0.5 V/ Div and adjust 4-6 div amplitude by using X 1 probe with the help of variable pot in Function Generator Block at input of Emitter.
8. Observe the output signal from detector on Oscilloscope.
9. Adjust the amplitude of the received signal as that of transmitted one with the help of gain adjusts potentiometer in AC amplifier block. Note this amplitude and name it V_1 .
10. Wind the fiber optic cable on the mandrel and observe the corresponding AC amplifier output on Oscilloscope, it will be gradually reducing, showing loss due to bends.

Observations: Observe the waveforms as shown below.



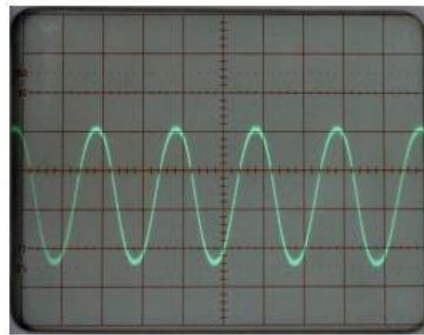
Input to Emitter Circuit

(CH1-0.2V/div; TB- 0.5mS)



Detector output (for 1m Cable)

(CH1-0.1V/div; TB- 0.5mS)



Detector output (after bending)

(CH1-0.1V/div; TB- 0.5mS)

Questions:

- What is the reason of bending losses?
- What is core and cladding?
- What is the function of cladding?